SUJAL GUPTA

Software Engineer | C++ & Systems Programming | AI/ML | Backend Engineering

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EDUCATION

Dr. B. R. Ambedkar National Institute of Technology, Jalandhar (NIT Jalandhar) — **Punjab, India**
Bachelor of Technology (B.Tech) in Mathematics & Computing (M&C) — *2024 – 2028 (Pre-final Year)*

- **Academic Standing:** Joint Entrance Examination (JEE) Main Common Rank List (CRL): **21,338** (Top ~1.3% among ~1.6 million candidates).
- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Discrete Mathematics, Linear Algebra, Probability & Statistics, Operating Systems, DBMS, Computer Networks, Theory of Computation.

TECHNICAL SKILLS

Languages	C++, Python, Java, Dart, C, SQL
AI & Machine Learning	Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Azure AI Search, Hugging Face Embeddings, ChromaDB, Vector Search, TensorFlow, LangGraph
Systems & Backend	FastAPI, REST APIs, Low-Latency Architecture, Multithreading, Memory Optimization, CMake, Linux / POSIX
Mobile & Frontend	Flutter, Dart, React (Learning), Responsive HTML5/CSS3
Databases & Cloud	PostgreSQL, Supabase, Firebase, Azure OpenAI, Git, GitHub, Docker

FEATURED PROJECTS

InvestIntel AI — Financial Intelligence & RAG Platform — **Python, FastAPI, Azure OpenAI, Azure AI Search**
AI-Powered Financial Document Analysis & Context-Aware Retrieval — *github.com/Sujal01go/AI-Invester-intelligence-platform*

- Architected an end-to-end financial document ingestion and chunking pipeline capable of processing 10-K, 10-Q filings and investor transcripts.
- Engineered hybrid retrieval combining semantic vector search (Azure AI Search & Hugging Face embeddings) with keyword ranking for zero-hallucination context extraction.
- Integrated GPT-4o context synthesis to accurately extract complex corporate financial KPIs, balance sheet metrics, and sentiment trends via high-throughput FastAPI endpoints.

Low-Latency Exchange Matching Engine — **C++20, STL, CMake, Microbenchmarking**
High-Performance Financial Exchange Order Matching Simulator — *Financial Technology & Systems*

- Implemented a high-throughput Limit Order Book (LOB) executing Price-Time Priority (FIFO) matching for buy and sell order flows.
- Optimized internal memory layout and cache locality using custom contiguous order structures and double-ended queues to minimize memory allocations on hot paths.
- Conducted microsecond-level latency profiling across 100,000+ synthetic order bursts, demonstrating sub-microsecond order insertion and execution latency.

AstroTrack — Mathematical Astronomy & Star Tracker — **Flutter, Dart, Coordinate Geometry, Sensors**
Offline-First Mobile Astronomy Engine & Constellation Tracker — *github.com/Sujal01go/star-tracker*

- Developed an offline astronomical catalog engine utilizing authentic Hipparcos and Yale Bright Star data to render real-time celestial maps.

- Engineered topocentric spherical trigonometry transformations to accurately compute local Altitude and Azimuth coordinates from J2000.0 epoch in real-time.
- Integrated hardware gyroscope and accelerometer orientation tracking for augmented reality (AR) night sky alignment.

Autonomous Multi-Agent Software Engineering Team **LangGraph, Python, Switch, SQLite**
Flip the Switch Hackathon *github.com/Sujal01go/switch-hackathon*

- Constructed an autonomous multi-agent developer system featuring Planner, Coder, Reviewer, and QA agents communicating over Telegram via Switch protocol.
- Leveraged LangGraph for state-machine orchestration and SQLite checkpointers for persistent fault-tolerant execution.

Sanjeevani — Ayurvedic Herb Intelligence Platform **Flutter, Dart, Supabase**
Herbal Wellness & Direct Farmer Marketplace Concept *github.com/Sujal01go/sanjeevani*

- Built a mobile application mapping symptoms to traditional medicinal herbs with real-time database queries on Supabase.
- Designed a verified farmer-to-consumer marketplace interface providing transparent pricing and verified medicinal plant listings.

EXPERIENCE & COMMUNITY INVOLVEMENT

GirlScript Summer of Code (GSSoC) **Remote**
Open Source Contributor *2026*

- Selected for national open-source program; contributed to open-source software repositories, resolving issues and submitting pull requests.

Utkansh Fest — NIT Jalandhar **Jalandhar, Punjab**
Public Relations & Event Organizing Team *2025 – 2026*

- Coordinated campus-wide and inter-collegiate outreach, managing public relations and communications for the institute's flagship annual cultural festival.

ACHIEVEMENTS & ACTIVITIES

- **LeetCode:** Solved 200+ algorithmic problems spanning Trees, Dynamic Programming, Graphs, and Advanced Data Structures.
- **JEE Main:** All India Common Rank List (CRL) 21,338 out of ~1.6 million candidates (98.7th percentile).
- **Hackathons:** Competed in HackTU (Thapar Institute), BIS Innovation Hackathon, Google Prompt Wars, and Flip the Switch Hackathon.
- **Bootcamps:** Google Developer Groups on Campus (GDGC) Flutter Bootcamp participant.